

Passband Digital Transmission

Binary and M-ary Digital Modulation

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Scope of this lecture

Binary modulation

- BPSK — signal space, detection, P_e
- BFSK — orthogonality, detection, P_e
- BASK (on-off keying)
- Transmitter and receiver structures

M-ary modulation

- MPSK and QPSK in detail
- MQAM
- MFSK
- Error performance and comparison

The single idea that organises everything

Every modulation scheme becomes a **set of points in a signal space**. Detection is then “pick the nearest point”, and performance is governed almost entirely by one number — the **minimum distance d_{min}** between points, measured against the noise.

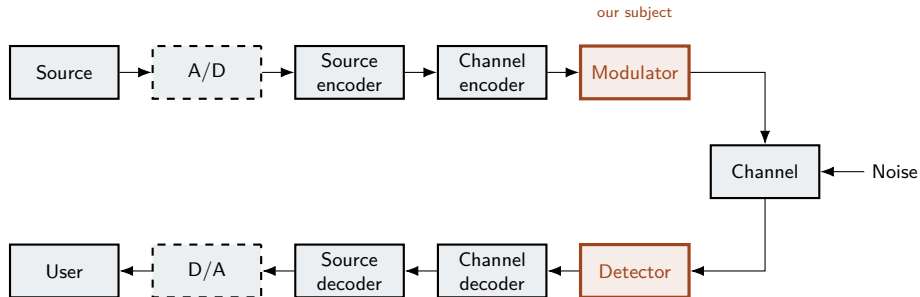
Outline

- 1 The Passband Transmission Problem
- 2 Binary Modulation
- 3 From Binary to M-ary
- 4 M-ary Phase-Shift Keying
- 5 M-ary Quadrature Amplitude Modulation
- 6 M-ary Frequency-Shift Keying
- 7 Comparison and Summary

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Where modulation sits in the system



- The **modulator** maps bits onto waveforms a physical channel can carry.
- The **detector** recovers those bits in the presence of additive noise.
- Everything between them — the channel — we model as **AWGN**: the received signal is the transmitted one plus white Gaussian noise of two-sided power spectral density $N_0/2$.

Why passband at all?

Baseband pulses occupy frequencies down to dc. Three reasons force us up to a carrier:

- 1 **Antenna size.** An efficient antenna is a sizeable fraction of a wavelength. At 3 kHz, $\lambda = 100$ km; at 3 GHz, $\lambda = 10$ cm.
- 2 **Channel suitability.** Most physical media (radio, optical fibre, satellite links) simply do not pass dc.
- 3 **Multiplexing.** Many users can share one medium only if each is translated to its own frequency band.

What modulation means for *digital* data

Modulation is the **switching** or **keying** of the amplitude, frequency or phase of a sinusoidal carrier according to the incoming bits:

$$s(t) = A(t) \cos(2\pi f_c t + \theta(t))$$

keying $A \rightarrow$ **ASK** (digital AM) keying $f_c \rightarrow$ **FSK** (digital FM) keying $\theta \rightarrow$ **PSK** (digital PM)

The signal-space approach

- Any finite set of M waveforms $\{s_m(t)\}$ can be written as a combination of at most M **orthonormal basis functions** $\{\phi_1, \dots, \phi_N\}$:

$$s_m(t) = \sum_{n=1}^N s_{mn} \phi_n(t), \quad s_{mn} = \int_0^T s_m(t) \phi_n(t) dt$$

- Each waveform collapses to a **point** (vector)
 $\mathbf{s}_m = [s_{m1}, \dots, s_{mN}]$.
- Energy becomes squared length:
 $E_m = \int_0^T s_m^2(t) dt = \|\mathbf{s}_m\|^2$.

Why this is worth the trouble

In AWGN the noise is **spherically symmetric** in this space. So:

- the optimum receiver is simply “choose the **nearest** signal point”;
- the error probability depends only on the **distances between points**, not on the waveform shapes.

Comparing modulation schemes reduces to comparing geometry.

Notation and the two energy symbols

Symbol	Meaning	Relation
T_b	bit duration	$R_b = 1/T_b$ bits/s
T	symbol duration	$T = kT_b$
M	number of waveforms	$M = 2^k$
k	bits carried per symbol	$k = \log_2 M$
E_b	energy <i>per bit</i>	—
E_s	energy <i>per symbol</i>	$E_s = kE_b$
N_0	noise PSD (one-sided)	two-sided $N_0/2$

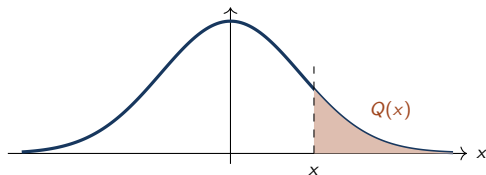
A constant source of confusion — settle it now

Schemes carrying different numbers of bits per symbol must be compared at equal E_b/N_0 , never at equal E_s/N_0 . Quoting E_s flatters large constellations, because one symbol of 16-QAM carries four bits' worth of energy.

The Q -function — and its relation to erfc

For a standard Gaussian random variable,

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^{\infty} e^{-u^2/2} du$$



Conversion (both appear in textbooks)

$$Q(x) = \frac{1}{2} \text{erfc}\left(\frac{x}{\sqrt{2}}\right), \quad \text{erfc}(x) = 2 Q(x\sqrt{2})$$

so, for example,

$$Q\left(\sqrt{\frac{2E_b}{N_0}}\right) = \frac{1}{2} \text{erfc}\left(\sqrt{\frac{E_b}{N_0}}\right)$$

Both forms give the same number — do not be misled into thinking two different results are being quoted.

The master result for two signals in AWGN

Binary detection in AWGN

If two equally likely signals $\mathbf{s}_1, \mathbf{s}_2$ are separated by Euclidean distance $d_{12} = \|\mathbf{s}_1 - \mathbf{s}_2\|$, the optimum (minimum-distance) receiver has error probability

$$P_e = Q\left(\frac{d_{12}}{\sqrt{2N_0}}\right)$$

- This one formula produces **every binary result** in this lecture. All that changes from scheme to scheme is d_{12} .
- It is also the building block for the M-ary case, where the **union bound** sums such pairwise terms and the **nearest neighbours** dominate.

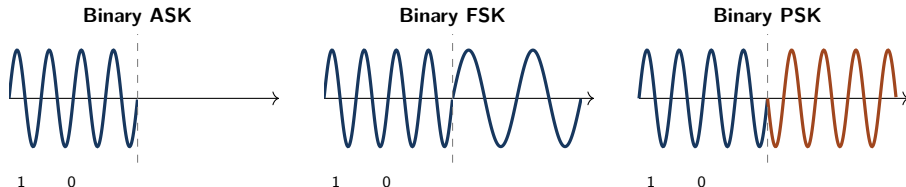
Design a modulation scheme = place M points so that d_{min} is as large as possible for a given average energy.

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Binary modulation: three ways to key a carrier

In binary signalling the modulator emits **one of two waveforms** per bit interval T_b .



- **BASK:** the carrier is switched on and off (on-off keying).
- **BFSK:** two different frequencies.
- **BPSK:** one frequency, two phases 180° apart.

Binary PSK: the signal set

$$\text{"1"} \rightarrow s_1(t) = \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t)$$

$$\begin{aligned}\text{"0"} \rightarrow s_2(t) &= \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t + \pi) \\ &= -\sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t)\end{aligned}$$

for $0 \leq t < T_b$.

- f_c is chosen as n_c/T_b for an integer n_c , or simply $f_c \gg 1/T_b$, so that each bit contains a whole number of cycles.
- E_b is the **energy per bit**:

$$\int_0^{T_b} s_1^2(t) dt = \int_0^{T_b} s_2^2(t) dt = E_b$$

- The two waveforms are identical except for a 180° phase reversal.

Note the structure

$s_2(t) = -s_1(t)$. The two signals are **antipodal** — as far apart as two signals of a given energy can possibly be. That is exactly why BPSK is the best-performing binary scheme.

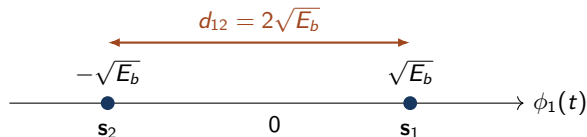
BPSK: signal space

There is **one** basis function:

$$\phi_1(t) = \sqrt{\frac{2}{T_b}} \cos(2\pi f_c t), \quad 0 \leq t < T_b$$

so that

$$s_1(t) = +\sqrt{E_b} \phi_1(t), \quad s_2(t) = -\sqrt{E_b} \phi_1(t)$$



Geometry

$N = 1$ dimension, $M = 2$ points,

$$d_{12} = 2\sqrt{E_b}$$

BPSK: decision rule

With the two symbols equally likely, $P(s_1) = P(s_2) = 0.5$, the optimum decision boundary is the **perpendicular bisector** of the line joining the two points — here simply the origin.

Decision rule

Form the correlator output r . Then

$$r > 0 \Rightarrow \text{decide } s_1 \text{ (bit 1)}$$

$$r \leq 0 \Rightarrow \text{decide } s_2 \text{ (bit 0)}$$

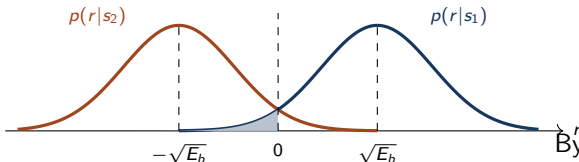


The received point is $r = \pm\sqrt{E_b} + n$, where n is a Gaussian random variable of zero mean and variance $N_0/2$. An error occurs when the noise carries r across the boundary.

BPSK: probability of error

Given s_1 was sent, an error occurs if $r < 0$:

$$\begin{aligned} P(e|s_1) &= \int_{-\infty}^0 \frac{1}{\sqrt{\pi N_0}} \exp\left\{-\frac{(r - \sqrt{E_b})^2}{N_0}\right\} dr \\ &= Q\left(\sqrt{\frac{2E_b}{N_0}}\right) \end{aligned}$$



By symmetry $P(e|s_2)$ is the same, so

$$P_e = Q\left(\sqrt{\frac{2E_b}{N_0}}\right) = \frac{1}{2} \operatorname{erfc}\left(\sqrt{\frac{E_b}{N_0}}\right)$$

Check against the master formula: $d_{12} = 2\sqrt{E_b}$, so $P_e = Q(d_{12}/\sqrt{2N_0}) = Q(\sqrt{2E_b/N_0})$. ✓

BPSK: what the result tells us

P_e depends on one quantity only

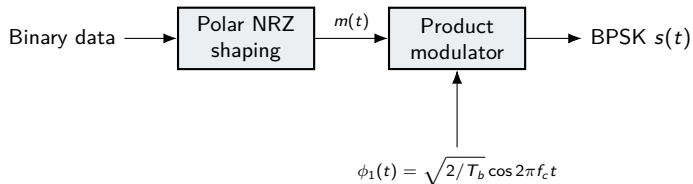
$$P_e = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$$

Not on f_c , not on T_b , not on the transmit power by itself — only on the ratio E_b/N_0 , the **bit energy to noise density ratio** (often called SNR per bit).

- Doubling the data rate at fixed power halves E_b , and so costs 3 dB.
- E_b/N_0 is therefore **the** currency of digital communications: it lets us compare systems of different rates and bandwidths fairly.

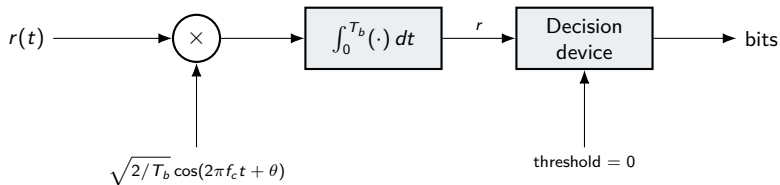
E_b/N_0	P_e
0 dB	7.9×10^{-2}
5 dB	6.0×10^{-3}
9.6 dB	1.0×10^{-5}
12 dB	9.0×10^{-9}

BPSK transmitter



- The bit stream is first mapped to a **polar NRZ** waveform: a “1” becomes $+\sqrt{E_b}$, a “0” becomes $-\sqrt{E_b}$.
- The product modulator multiplies that waveform by the basis function $\phi_1(t)$ — multiplying by ± 1 is exactly what produces the 180° phase reversal.
- The transmitted power is constant: BPSK has a **constant envelope**.

BPSK receiver (coherent detection)



Why “coherent”

The local carrier must match the incoming one in both **phase** and **timing**. The offset θ arises from propagation delay and from transmitter and receiver oscillators that are not synchronous. A carrier-recovery loop must estimate it, because the correlator output is scaled by $\cos \theta$ - a 60° phase error halves it, and 90° destroys it entirely.

Binary FSK: the signal set

$$\text{"1"} \rightarrow s_1(t) = \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_1 t)$$

$$\text{"0"} \rightarrow s_2(t) = \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_2 t)$$

for $0 \leq t < T_b$, with $\int_0^{T_b} s_i^2(t) dt = E_b$.

The orthogonality requirement

The two frequencies must be chosen so that

$$\int_0^{T_b} s_1(t)s_2(t) dt = 0$$

Carrying out the integral gives $\text{sinc}(2\Delta f T_b) = 0$, whose first zero is at

$$\Delta f = f_1 - f_2 = \frac{1}{2T_b}$$

Why orthogonality matters

Orthogonal signals are **perpendicular** in signal space. Antipodal signals (BPSK) point in opposite directions. For the same energy, perpendicular points are closer together than opposite ones — and that gap is precisely the 3 dB penalty we are about to derive.

BFSK: signal space

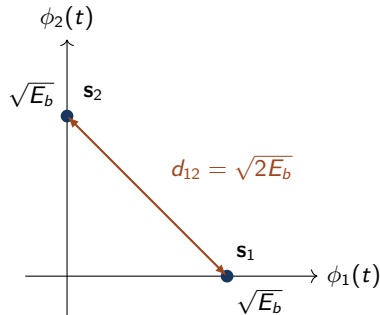
Two basis functions are needed:

$$\phi_1(t) = \sqrt{\frac{2}{T_b}} \cos(2\pi f_1 t)$$

$$\phi_2(t) = \sqrt{\frac{2}{T_b}} \cos(2\pi f_2 t)$$

so that $s_1(t) = \sqrt{E_b} \phi_1(t)$, $s_2(t) = \sqrt{E_b} \phi_2(t)$, i.e.

$$\mathbf{s}_1 = [\sqrt{E_b} \ 0], \quad \mathbf{s}_2 = [0 \ \sqrt{E_b}]$$

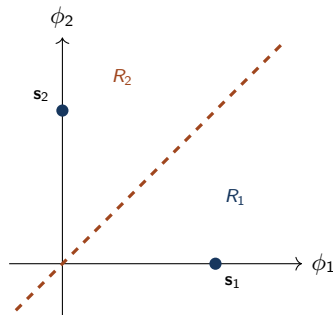


Geometry

$N = 2$ dimensions, $M = 2$ points,

$$d_{12} = \sqrt{2E_b}$$

BFSK: decision regions and error probability



The receiver computes $r_1 = \int_0^{T_b} r(t)\phi_1 dt$ and $r_2 = \int_0^{T_b} r(t)\phi_2 dt$, then compares them: decide $\mathbf{s}_1$ if $r_1 > r_2$.

Given $\mathbf{s}_1$ sent: $r_1 = \sqrt{E_b} + n_1$, $r_2 = n_2$, so

$$P(e|s_1) = P(\sqrt{E_b} + n_1 < n_2) = P(n < -\sqrt{E_b})$$

where $n = n_1 - n_2$. Since n_1, n_2 are i.i.d. $\mathcal{N}(0, N_0/2)$, $n \sim \mathcal{N}(0, N_0)$. Hence

$$P_e = Q\left(\sqrt{\frac{E_b}{N_0}}\right)$$

Again consistent with the master formula: $d_{12} = \sqrt{2E_b} \Rightarrow P_e = Q(\sqrt{2E_b}/\sqrt{2N_0}) = Q(\sqrt{E_b/N_0})$. ✓

BFSK versus BPSK: the 3 dB penalty

The two results side by side

$$P_e^{\text{BPSK}} = Q\left(\sqrt{\frac{2E_b}{N_0}}\right), \quad P_e^{\text{BFSK}} = Q\left(\sqrt{\frac{E_b}{N_0}}\right)$$

BFSK needs **twice the energy** (3 dB more power) to reach the same P_e .

Where the 3 dB comes from

It is pure geometry:

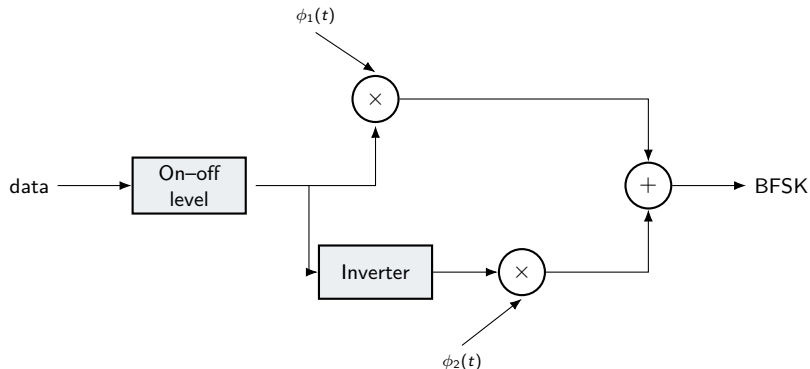
$$d_{12}^{\text{BPSK}} = 2\sqrt{E_b}, \quad d_{12}^{\text{BFSK}} = \sqrt{2E_b}$$

$$\frac{d^{\text{BPSK}}}{d^{\text{BFSK}}} = \frac{2\sqrt{E_b}}{\sqrt{2E_b}} = \sqrt{2}$$

A distance ratio of $\sqrt{2}$ is an energy ratio of 2, i.e. **3 dB**.

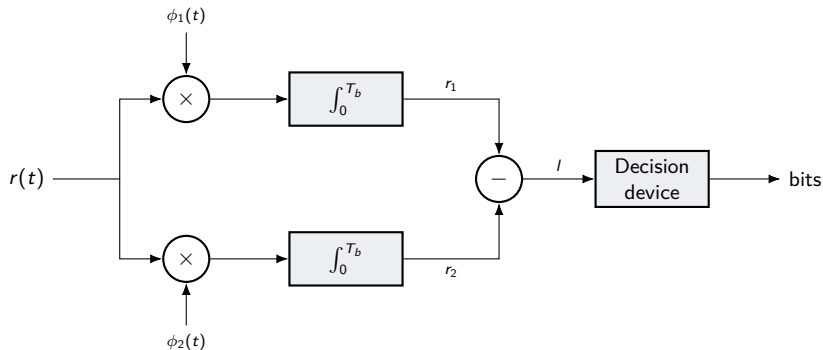
Antipodal signalling beats orthogonal signalling by 3 dB — always, and for this reason alone. Nothing about frequencies or phases enters the argument.

BFSK transmitter



- The data is put into **on-off** form, so exactly one of the two branches is active at any moment.
- A “1” enables the f_1 branch; the inverter enables the f_2 branch for a “0”.
- The adder combines them — but only one is ever non-zero, so the output is a single tone per bit.

Coherent BFSK receiver



- Two correlators measure how much of each tone is present, giving r_1 and r_2 .
- The decision is a **comparison**, not a threshold test: decide 1 if $I = r_1 - r_2 > 0$, else 0.
- Because it is a comparison, the result is **independent of the received amplitude** — unlike BASK.

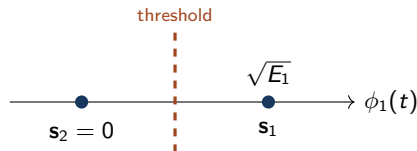
Binary ASK (on-off keying)

$$\text{"1"} \rightarrow s_1(t) = \sqrt{\frac{2E_1}{T_b}} \cos(2\pi f_c t), \quad \text{"0"} \rightarrow s_2(t) = 0$$

- Only **one** basis function, so $N = 1$ — but unlike BPSK the two points are **not symmetric about the origin**.
- Since nothing is sent for a "0", the *average energy per bit* is $E_b = E_1/2$.
- $d_{12} = \sqrt{E_1} = \sqrt{2E_b}$, hence

$$P_e = Q\left(\sqrt{\frac{E_b}{N_0}}\right)$$

— identical to BFSK.



Practical drawback

The optimum threshold sits at $\sqrt{E_1}/2$, which **depends on the received amplitude**. Any fading or gain drift moves it. BPSK's threshold is zero and needs no such calibration.

Binary schemes compared

	BPSK	BFSK	BASK
Dimensions N	1	2	1
Geometry	antipodal	orthogonal	unipolar
Distance d_{12}	$2\sqrt{E_b}$	$\sqrt{2E_b}$	$\sqrt{2E_b}$
P_e	$Q\left(\sqrt{2E_b/N_0}\right)$	$Q\left(\sqrt{E_b/N_0}\right)$	$Q\left(\sqrt{E_b/N_0}\right)$
E_b/N_0 for 10^{-5}	9.6 dB	12.6 dB	12.6 dB
Threshold	0, fixed	compare r_1, r_2	amplitude dependent
Envelope	constant	constant	not constant

Summary of the binary story

All three results come from one formula, $P_e = Q(d_{12}/\sqrt{2N_0})$. BPSK wins by 3 dB purely because antipodal points are $\sqrt{2}$ further apart than orthogonal ones of the same energy.

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The M-ary idea

- In binary signalling we send one of **two** waveforms every T_b seconds.
- In M-ary signalling we send one of **M** waveforms every T seconds. Each such waveform is called a **symbol**.
- In almost all practical systems

$$M = 2^k, \quad T = kT_b, \quad k = \log_2 M$$

so one symbol carries k bits.

The symbols are made by keying the carrier in M discrete steps

MASK (amplitude) **MPSK** (phase) **MFSK** (frequency) **MQAM** (amplitude *and* phase)

Why bother

The symbol rate is $R_s = R_b/k$. Transmitting k bits per symbol lets us send the same bit rate in roughly **$1/k$ of the bandwidth** — the central reason modern systems use large constellations.

The trade-off that governs all M-ary design

Bandwidth efficiency

$$\rho = \frac{R_b}{B} \text{ bits/s/Hz}$$

Packing more bits into a symbol **increases** ρ .

Power efficiency

The E_b/N_0 needed for a target P_e . Packing more points into the same space makes them **closer**, so more power is needed.

Two opposite families

MPSK and MQAM are **bandwidth-efficient**:

$N = 2$ dimensions regardless of M , so bandwidth stays fixed while rate rises — but d_{min} shrinks and power efficiency falls.

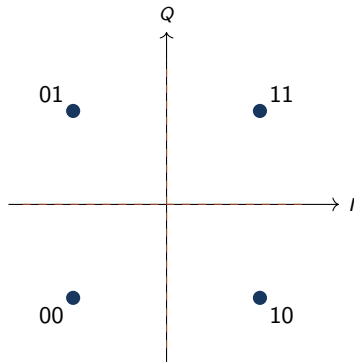
MFSK is **power-efficient**: it uses M orthogonal dimensions, so d_{min} is maintained as M grows — but bandwidth grows with M .

You cannot have both. Choosing a modulation is choosing where on this trade-off to sit.

Symbol errors, bit errors and Gray coding

- Analysis naturally yields the **symbol** error probability P_s . What we care about is the **bit** error probability P_b .
- At useful SNRs an error almost always lands on an **adjacent** constellation point — never a distant one.
- **Gray coding** assigns bit patterns so that adjacent points differ in **exactly one bit**. A symbol error then causes just one bit error:

$$P_b \approx \frac{P_s}{k}$$



Gray-mapped QPSK: every neighbour differs in one bit.

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MPSK: the signal set

The carrier phase takes one of M equally spaced values:

$$\theta_m = \frac{2\pi(m-1)}{M}, \quad m = 1, \dots, M$$

$$s_m(t) = \sqrt{\frac{2E_s}{T}} \cos\left[2\pi f_c t + \frac{2\pi(m-1)}{M}\right], \quad 0 \leq t < T$$

- E_s is the energy *per symbol*; $f_c \gg 1/T$.
- Every symbol has the **same amplitude** — MPSK is a **constant-envelope** modulation.

Basis functions

$$\phi_1(t) = \sqrt{\frac{2}{T}} \cos(2\pi f_c t)$$

$$\phi_2(t) = \sqrt{\frac{2}{T}} \sin(2\pi f_c t)$$

Constant envelope matters in practice: it allows the transmitter's power amplifier to be driven into saturation (class C or near it) without distorting the signal — a decisive advantage in satellite and handset transmitters.

MPSK: signal space representation

Expanding the cosine of a sum,

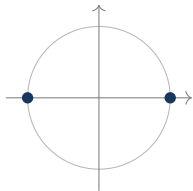
$$\begin{aligned}s_m(t) &= \sqrt{\frac{2E_s}{T}} \cos(2\pi f_c t) \cos\left[\frac{2\pi(m-1)}{M}\right] - \sqrt{\frac{2E_s}{T}} \sin(2\pi f_c t) \sin\left[\frac{2\pi(m-1)}{M}\right] \\ &= \sqrt{E_s} \cos\left[\frac{2\pi(m-1)}{M}\right] \phi_1(t) - \sqrt{E_s} \sin\left[\frac{2\pi(m-1)}{M}\right] \phi_2(t)\end{aligned}$$

so the signal vector is

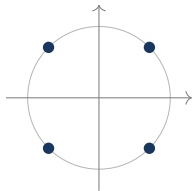
$$\mathbf{s}_m = \left[\sqrt{E_s} \cos \frac{2\pi(m-1)}{M} \quad \sqrt{E_s} \sin \frac{2\pi(m-1)}{M} \right], \quad m = 1, \dots, M$$

All M points lie on a **circle of radius $\sqrt{E_s}$** — a direct restatement of the constant-envelope property. Only two dimensions are used no matter how large M becomes.

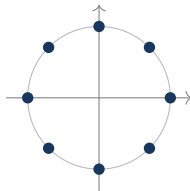
MPSK constellations



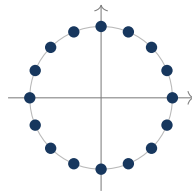
BPSK, $M = 2$



QPSK, $M = 4$



8-PSK, $M = 8$



16-PSK, $M = 16$

What to notice

The circle's radius $\sqrt{E_s}$ is fixed by the transmit power, but the points crowd together as M grows.

Doubling M roughly halves the spacing — and error performance follows the spacing, not the number of points.

MPSK: minimum distance and error probability

The distance between any two points is

$$d_{mn} = \|\mathbf{s}_m - \mathbf{s}_n\| = \sqrt{2E_s \left(1 - \cos \frac{2\pi(m-n)}{M} \right)}$$

The closest pair are adjacent ($|m - n| = 1$), giving

$$d_{min} = \sqrt{2E_s \left(1 - \cos \frac{2\pi}{M} \right)} = 2\sqrt{E_s} \sin \frac{\pi}{M}$$

Errors are dominated by confusion with the **two adjacent** points, so applying the master formula twice,

$$P_s \approx 2 Q \left(\frac{d_{min}/2}{\sqrt{N_0}/2} \right) = 2 Q \left(\sqrt{\frac{2E_s}{N_0}} \sin \frac{\pi}{M} \right)$$

and with Gray coding $P_b \approx P_s/k$, where $E_s = kE_b$.

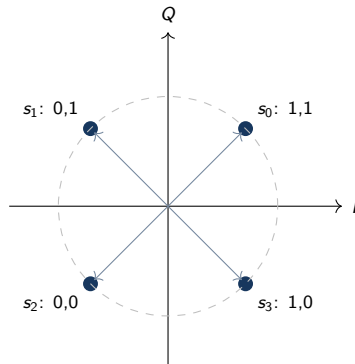
Quadrature PSK: the important special case

With $M = 4$ ($k = 2$ bits per symbol) the four phases are usually placed at $45^\circ, 135^\circ, 225^\circ, 315^\circ$:

$$s_i(t) = A \cos(2\pi f_c t + \phi_i), \quad \phi_i = (2i + 1)\frac{\pi}{4}$$

for $i = 0, 1, 2, 3$.

- Horizontal axis: **in-phase (I)** component.
- Vertical axis: **quadrature (Q)** component, 90° shifted.



The key structural fact

Writing $s_i(t) = A_I \cos \omega_0 t + A_Q \sin \omega_0 t$ shows QPSK to be **two independent BPSK signals** riding on orthogonal carriers. Everything about QPSK follows from this.

QPSK: I and Q components

i	ϕ_i	A_I	A_Q	Binary data
0	45°	+1	+1	1, 1
1	135°	-1	+1	0, 1
2	-135°	-1	-1	0, 0
3	-45°	+1	-1	1, 0

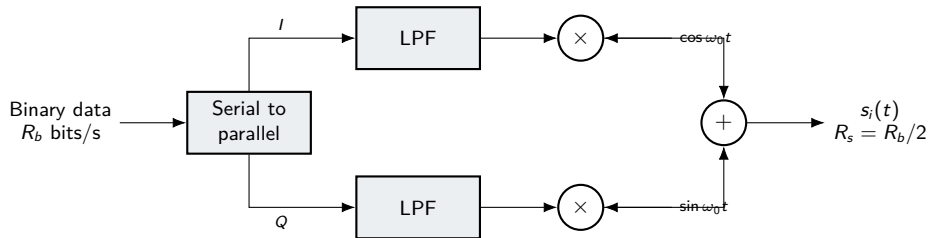
Read the table as two BPSK streams

The **first bit** sets the sign of A_I ; the **second bit** sets the sign of A_Q . The two are completely independent.

This mapping is Gray

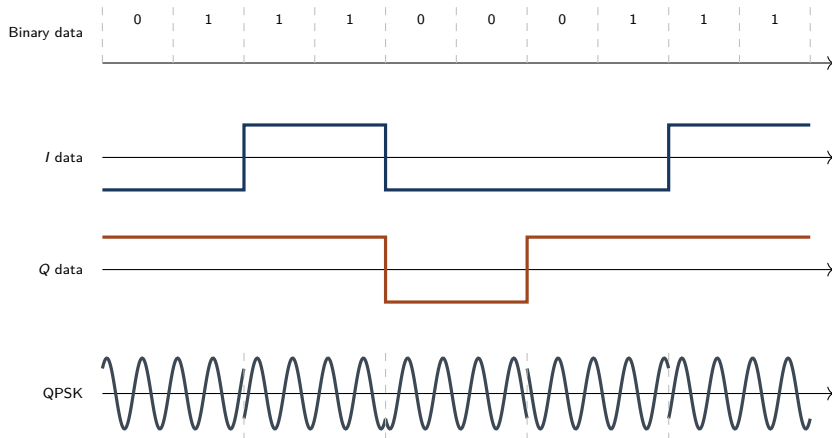
Adjacent phases differ in exactly one bit, so a symbol error — which almost always goes to a neighbour — costs only **one** bit error, not two. The assignment is not unique, but this choice is the one worth making.

QPSK modulator



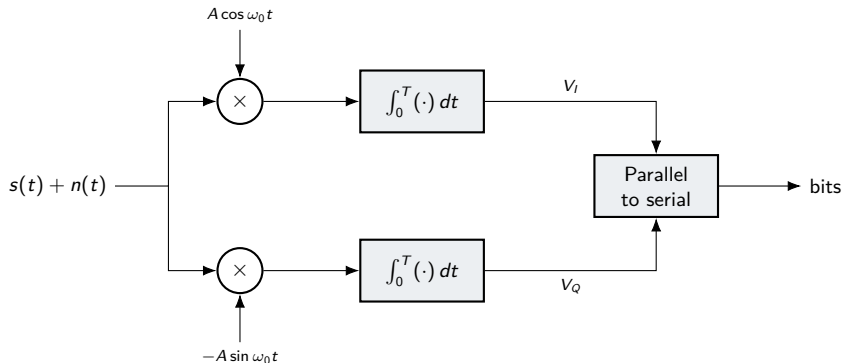
- The serial bit stream is split into **pairs**: odd bits drive I , even bits drive Q .
- A 90° hybrid derives both carriers from one oscillator.
- Because the symbol rate is halved, **the QPSK spectrum is half as wide as BPSK at the same bit rate.**

QPSK: waveforms



Each symbol lasts $T = 2T_b$ and the phase can change only at symbol boundaries.

QPSK demodulator



- Demodulation requires **coherent detection**: two mixers driven by quadrature local-oscillator components recover I and Q.
- Each branch is an integrate-and-dump detector sampled at $t = T$.
- Because QPSK has a **constant envelope**, both detectors use a **zero threshold** — no amplitude calibration is needed.

QPSK: correlator outputs

With $s_i(t) = A \cos(\omega_0 t + \phi_i)$ received, the I correlator gives

$$\begin{aligned} V_I &= \int_0^T A^2 \cos(\omega_0 t + \phi_i) \cos \omega_0 t \, dt \\ &= \frac{A^2}{2} \int_0^T [\cos(2\omega_0 t + \phi_i) + \cos \phi_i] \, dt = \frac{A^2 T}{2} \cos \phi_i \end{aligned}$$

and similarly $V_Q = \frac{A^2 T}{2} \sin \phi_i$.

$s_i(t)$	ϕ_i	V_I	V_Q	Binary output
s_0	45°	$+C$	$+C$	1, 1
s_1	135°	$-C$	$+C$	0, 1
s_2	-135°	$-C$	$-C$	0, 0
s_3	-45°	$+C$	$-C$	1, 0

where $C = A^2 T / 2\sqrt{2}$. The **signs** of V_I and V_Q recover the two bits directly.

QPSK: probability of error

By symmetry take s_1 (at 135°) as transmitted. With $E_s = A^2 T/2$,

$$V_I(T) = -\frac{E_s}{\sqrt{2}} + n_o(T), \quad V_Q(T) = +\frac{E_s}{\sqrt{2}} + n_o(T)$$

where the noise term has variance $\sigma^2 = E_s N_0/2$.

Each detector has a **zero threshold**, so the error probability at the I correlator is

$$P_e^{(I)} = P\{V_I > 0\} = \frac{1}{2} \operatorname{erfc}\left(\sqrt{\frac{E_s}{2N_0}}\right) = Q\left(\sqrt{\frac{E_s}{N_0}}\right)$$

and by symmetry $P_e^{(Q)}$ is identical. A symbol is correct only if *both* correlators are correct:

$$P_s = 1 - (1 - P_e^{(I)})(1 - P_e^{(Q)}) \approx 2P_e^{(I)} = \operatorname{erfc}\left(\sqrt{\frac{E_s}{2N_0}}\right) \quad (P_e \ll 1)$$

QPSK: the headline result

With Gray coding a symbol error causes one bit error, and each symbol carries two bits, so

$$P_b = \frac{P_s}{2} = \frac{1}{2} \operatorname{erfc} \left(\sqrt{\frac{E_s}{2N_0}} \right)$$

Now substitute $E_s = 2E_b$ (the symbol lasts two bit periods):

$$P_b = \frac{1}{2} \operatorname{erfc} \left(\sqrt{\frac{E_b}{N_0}} \right) = Q \left(\sqrt{\frac{2E_b}{N_0}} \right)$$

Identical to BPSK — and this is the point

QPSK achieves **exactly the same bit error rate as BPSK** while carrying **twice the data in the same bandwidth**. There is no penalty whatsoever.

The reason is the structural fact noted earlier: QPSK is two orthogonal BPSK channels. Orthogonal channels do not interfere, so each simply performs as BPSK. This is why QPSK, not BPSK, is the workhorse of satellite and cellular links.

Worked example: QPSK versus BPSK on a radio link

Problem

A QPSK system transmits at 20 Mbps over a radio link with 2 W transmit power. The total link loss is 110 dB and the two-sided noise PSD is $N_0/2 = 4 \times 10^{-20}$ W/Hz. With Gray coding, find the bit error rate and compare with BPSK.

Received power

$$P_r = 2 \times 10^{-110/10} = 2 \times 10^{-11} \text{ W}$$

QPSK — symbol rate is half the bit rate, so $E_s = P_r/R_s = 2P_r/R_b$:

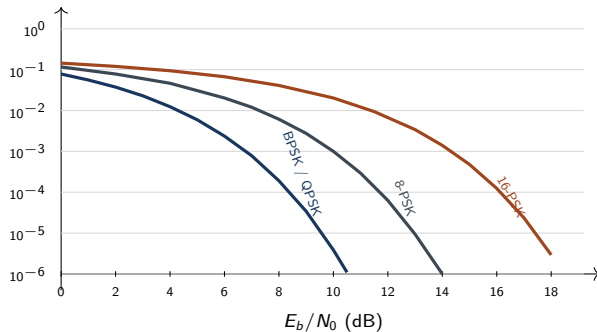
$$\frac{E_s}{N_0} = \frac{2(2 \times 10^{-11})}{(8 \times 10^{-20})(20 \times 10^6)} = 25 \quad \Rightarrow \quad P_b = \frac{1}{2}\text{erfc}\left(\sqrt{\frac{25}{2}}\right) = \mathbf{2.87 \times 10^{-7}}$$

BPSK — $E_b/N_0 = P_r/(N_0R_b) = 12.5$:

$$P_b = \frac{1}{2}\text{erfc}(\sqrt{12.5}) = \mathbf{2.87 \times 10^{-7}}$$

Identical — as predicted.

Error performance of MPSK



- $M = 2$ and $M = 4$ give identical bit error rates.
- Beyond that, every doubling of M costs power. For large M ,

$$d_{min} = 2\sqrt{E_s} \sin \frac{\pi}{M} \approx \frac{2\pi\sqrt{E_s}}{M}$$

so halving the spacing requires roughly **6 dB more energy per bit** for each extra bit per symbol.

E_b/N_0 for $P_b = 10^{-5}$

BPSK/QPSK 9.6 dB 8-PSK 13.0 dB
16-PSK 17.5 dB

Exercise: distances in PSK constellations

Problem

Consider $M = 2, 4, 8$ PSK constellations, all with the same symbol energy E_s .

- 1 Determine d_{min} in each case.
- 2 For $M = 8$, by how many dB must E_s be increased to achieve the same d_{min} as $M = 4$?

Solution. Using $d_{min} = 2\sqrt{E_s} \sin(\pi/M)$:

M	$\sin(\pi/M)$	d_{min}	relative to $M = 2$
2	1	$2\sqrt{E_s}$	1
4	0.7071	$1.414\sqrt{E_s}$	0.707
8	0.3827	$0.765\sqrt{E_s}$	0.383

For equal d_{min} we need $E_s \sin^2(\pi/M)$ equal, so

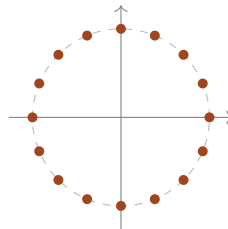
$$\frac{E_s^{(8)}}{E_s^{(4)}} = \frac{\sin^2(\pi/4)}{\sin^2(\pi/8)} = \frac{0.5}{0.1464} = 3.41 \quad \Rightarrow \quad 10 \log_{10} 3.41 = \mathbf{5.33 \text{ dB}}$$

Outline

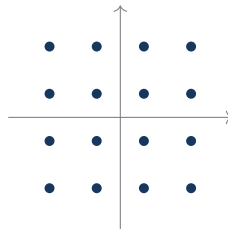
- 1 The Passband Transmission Problem
- 2 Binary Modulation
- 3 From Binary to M-ary
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- 5 M-ary Quadrature Amplitude Modulation**
- 6 M-ary Frequency-Shift Keying
- 7 Comparison and Summary

From MPSK to MQAM

- In MPSK the I and Q components are **constrained**: they must satisfy $a^2 + b^2 = E_s$, putting every point on a circle.
- That constraint wastes the interior of the plane. As M grows the points crowd onto the circumference.
- **Relax it**: let the amplitude vary as well as the phase. The result is **M-ary QAM** — also called APSK, amplitude and phase-shift keying.



16-PSK



16-QAM

Why QAM wins for large M

For the same *average* energy, spreading 16 points over a square lattice gives a substantially larger d_{min} than squeezing them onto a circle. **16-QAM outperforms 16-PSK by about 4 dB**. The price is that the envelope is no longer constant, so the power amplifier must stay linear.

MQAM: signal set and signal space

Signal set

$$s_i(t) = \sqrt{\frac{2E_0}{T}} a_i \cos(2\pi f_c t) + \sqrt{\frac{2E_0}{T}} b_i \sin(2\pi f_c t), \quad 0 \leq t < T$$

where E_0 is the energy of the lowest-amplitude signal and (a_i, b_i) is a pair of independent odd integers.

Basis functions

$$\phi_1(t) = \sqrt{\frac{2}{T}} \cos(2\pi f_c t)$$

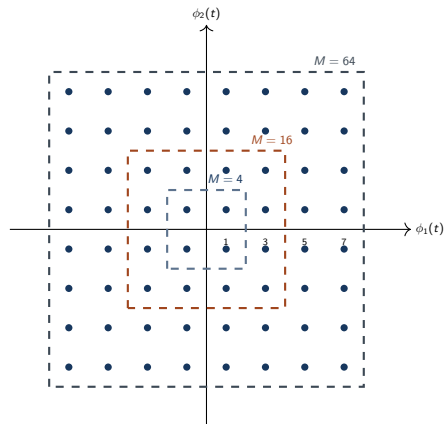
$$\phi_2(t) = \sqrt{\frac{2}{T}} \sin(2\pi f_c t)$$

Signal vector

$$\mathbf{s}_i = \begin{bmatrix} \sqrt{E_0} a_i & \sqrt{E_0} b_i \end{bmatrix}$$

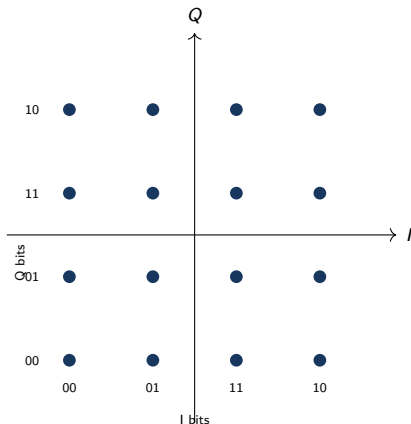
Still only **two dimensions**, exactly as MPSK — so MQAM occupies the same bandwidth for the same symbol rate.

MQAM constellations: the square lattice



- The co-ordinates are **odd integers** $\pm 1, \pm 3, \pm 5, \dots$ times $\sqrt{E_0}$, so neighbouring points are always $2\sqrt{E_0}$ apart: $d_{min} = 2\sqrt{E_0}$ regardless of M .
- Enlarging M therefore does not reduce the spacing — it increases the *average energy* instead. That is a far better bargain than MPSK offers.

16-QAM with Gray mapping



- Each point carries $k = 4$ bits: two set the I co-ordinate, two the Q.
- Both axes are **Gray coded independently**, so any move to a horizontal or vertical neighbour changes exactly one bit.
- Because I and Q are independent, 16-QAM is literally **two independent 4-ASK signals** on quadrature carriers — the same decomposition that made QPSK easy.

Practical note

64-QAM and 256-QAM are mandated for digital cable (ANSI/SCTE 07) and are standard in LTE, Wi-Fi and DVB.

MQAM: error performance

Upper bound on symbol error probability

$$P_s \leq 4 Q \left(\sqrt{\frac{3k E_b}{(M-1)N_0}} \right), \quad M = 2^k$$

and with Gray coding $P_b \approx P_s/k$.

Exercise

Determine the increase in E_b required to maintain the same error performance when the number of bits per symbol rises from k to $k+1$, for large k .

Solution. The argument of Q must be held constant:

$$\frac{3kE_b}{M-1} = \frac{3(k+1)E'_b}{2M-1} \implies \frac{E'_b}{E_b} = \frac{k}{k+1} \cdot \frac{2M-1}{M-1} \xrightarrow[k \text{ large}]{} 2$$

about 3 dB per extra bit per symbol — compare with roughly 6 dB for MPSK.

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MFSK (multitone signalling)

Signal set

$$s_m(t) = \sqrt{\frac{2E_s}{T}} \cos\{2\pi(f_c + (m-1)\Delta f)t\}, \quad m = 1, \dots, M, \quad 0 \leq t < T$$

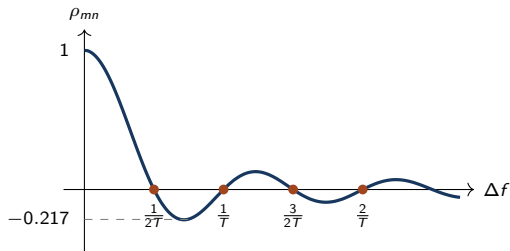
with tone spacing $\Delta f = f_m - f_{m-1}$.

The correlation between two of the tones is

$$\rho_{mn} = \frac{1}{E_s} \int_0^T s_m(t)s_n(t) dt = \frac{\sin[2\pi(m-n)\Delta f T]}{2\pi(m-n)\Delta f T} = \text{sinc}[2(m-n)\Delta f T]$$

Setting $\rho_{mn} = 0$ gives the **minimum spacing for orthogonality**.

MFSK: choosing the tone spacing



Minimum orthogonal spacing

$$\Delta f = \frac{1}{2T}$$

the first zero of the sinc.

- Total bandwidth is therefore roughly $B \approx M\Delta f = M/2T$ — it **grows linearly with M** .
- The minimum of $\rho_{mn} = -0.217$ at $\Delta f \approx 0.715/T$ would give slightly better performance, but the signals are then no longer orthogonal and the receiver is more complex.

MFSK: signal space and performance

With $\Delta f = 1/2T$ the M tones are mutually orthogonal, so each needs its own dimension. The basis functions are

$$\phi_m(t) = \sqrt{\frac{2}{T}} \cos 2\pi(f_c + m\Delta f)t$$

and the signal vectors are

$$\mathbf{s}_1 = (\sqrt{E_s}, 0, \dots, 0), \quad \mathbf{s}_2 = (0, \sqrt{E_s}, \dots, 0), \quad \dots$$

$$N = M \text{ dimensions, } d_{\min} = \sqrt{2E_s} \text{ for every } M.$$

The opposite trade to MPSK/MQAM

Because each extra symbol gets a **new orthogonal dimension**, the points never crowd: $d_{\min}$ is independent of M . Increasing M therefore **improves** power efficiency — at the cost of bandwidth growing as M .

Limit

As $M \rightarrow \infty$ the required E_b/N_0 approaches the **Shannon limit of -1.6 dB** — but the bandwidth becomes unbounded.

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Bits per symbol and symbol rate

Modulation	Bits/symbol k	Symbol rate	Dimensions
BPSK	1	$1 \times$ bit rate	1
QPSK	2	$1/2$ bit rate	2
8-PSK	3	$1/3$ bit rate	2
16-QAM	4	$1/4$ bit rate	2
32-QAM	5	$1/5$ bit rate	2
64-QAM	6	$1/6$ bit rate	2
M -FSK	$\log_2 M$	$1/\log_2 M$ bit rate	M

Read the last column

PSK and QAM always use **two** dimensions, so their bandwidth is set by the symbol rate alone — raising k buys bandwidth efficiency directly. MFSK uses **M** dimensions, which is why it behaves in the opposite way.

Master comparison

Scheme	Dim.	d_{min}	P_e (coherent)	E_b/N_0 @ 10^{-5}	Envelope
BPSK	1	$2\sqrt{E_b}$	$Q\left(\sqrt{2E_b/N_0}\right)$	9.6 dB	constant
BFSK	2	$\sqrt{2E_b}$	$Q\left(\sqrt{E_b/N_0}\right)$	12.6 dB	constant
BASK	1	$\sqrt{2E_b}$	$Q\left(\sqrt{E_b/N_0}\right)$	12.6 dB	varying
QPSK	2	$\sqrt{2E_s}$	$Q\left(\sqrt{2E_b/N_0}\right)$	9.6 dB	constant
MPSK	2	$2\sqrt{E_s} \sin \frac{\pi}{M}$	$2Q\left(\sqrt{2E_s/N_0} \sin \frac{\pi}{M}\right)$	+6 dB per bit	constant
MQAM	2	$2\sqrt{E_0}$	$\leq 4Q\left(\sqrt{\frac{3kE_b}{(M-1)N_0}}\right)$	+3 dB per bit	varying
MFSK	M	$\sqrt{2E_s}$	union bound	improves with M	constant

The one-line summary

Bandwidth-efficient → **MQAM**, then MPSK. Power-efficient → **MFSK**. Best all-round compromise → **QPSK**.

Formula sheet (1 of 2)

General

$$P_e = Q\left(\frac{d_{12}}{\sqrt{2N_0}}\right), \quad Q(x) = \frac{1}{2}\operatorname{erfc}\left(\frac{x}{\sqrt{2}}\right)$$

$$E_s = kE_b, \quad M = 2^k, \quad T = kT_b$$

Binary schemes

$$P_e^{\text{BPSK}} = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$$

$$P_e^{\text{BFSK}} = P_e^{\text{BASK}} = Q\left(\sqrt{\frac{E_b}{N_0}}\right)$$

$$\Delta f_{\text{BFSK}} = \frac{1}{2T_b}$$

MPSK / QPSK

$$\mathbf{s}_m = \sqrt{E_s} \left[\cos \frac{2\pi(m-1)}{M} \quad \sin \frac{2\pi(m-1)}{M} \right]$$

$$d_{\min} = 2\sqrt{E_s} \sin \frac{\pi}{M}$$

$$P_s \approx 2Q\left(\sqrt{\frac{2E_s}{N_0}} \sin \frac{\pi}{M}\right), \quad P_b \approx \frac{P_s}{k}$$

$$P_b^{\text{QPSK}} = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$$

Formula sheet (2 of 2)

MQAM

$$\mathbf{s}_i = \sqrt{E_0} [a_i \ b_i], \quad d_{min} = 2\sqrt{E_0}$$

$$P_s \leq 4Q\left(\sqrt{\frac{3kE_b}{(M-1)N_0}}\right)$$

≈ 3 dB cost per extra bit per symbol

MFSK

$$\rho_{mn} = \text{sinc}[2(m-n)\Delta f T]$$

$$\Delta f_{min} = \frac{1}{2T}, \quad B \approx \frac{M}{2T}$$

$N = M$ dimensions, $d_{min} = \sqrt{2E_s}$ for all M

Useful numerical values of $Q(x)$

x	1	2	3	4	5
$Q(x)$	1.59×10^{-1}	2.28×10^{-2}	1.35×10^{-3}	3.17×10^{-5}	2.87×10^{-7}

Tutorial problems

- ❶ A BPSK system operates with $E_b/N_0 = 8$ dB. Find P_e . Repeat for coherent BFSK. (1.9×10^{-4} , 1.0×10^{-2})
- ❷ A BPSK link must achieve $P_e = 10^{-6}$. Find the required E_b/N_0 in dB. (*10.5 dB*)
- ❸ Binary data at 1 Mbps is sent by BFSK with $f_1 = 10.1$ MHz. Find the smallest f_2 giving orthogonality. (*10.6 MHz*)
- ❹ For 8-PSK with $E_s/N_0 = 14$ dB, estimate the symbol error probability and, assuming Gray coding, the bit error probability. (3.5×10^{-3} , 1.2×10^{-3})
- ❺ A QPSK system transmits 50 Mbps. What symbol rate is required, and how does the bandwidth compare with BPSK at the same bit rate? (*25 Mbaud, half*)
- ❻ Show that 16-QAM requires about 4 dB less E_b/N_0 than 16-PSK for the same symbol error probability.
- ❼ An MFSK system uses $M = 8$ with $T = 1$ ms. Find the minimum tone spacing and the approximate transmission bandwidth. (*500 Hz, 4 kHz*)
- ❽ Explain why QPSK carries twice the data of BPSK in the same bandwidth with no loss in bit error rate.

Key take-aways

- 1 Passband modulation keys the **amplitude, frequency or phase** of a carrier. Every scheme reduces to a **constellation of points**.
- 2 In AWGN the optimum receiver chooses the **nearest point**, and $P_e = Q(d/\sqrt{2N_0})$ for a pair of signals. Everything else is geometry.
- 3 **Antipodal beats orthogonal by 3 dB**: BPSK needs 9.6 dB for 10^{-5} , BFSK and BASK need 12.6 dB.
- 4 **QPSK is BPSK twice over** — two orthogonal carriers, twice the rate, same bandwidth, same BER. This is why it is used everywhere.
- 5 Going to larger M in **PSK or QAM** buys bandwidth efficiency and costs power (≈ 6 dB per bit for PSK, ≈ 3 dB for QAM). Going to larger M in **FSK** does the reverse.
- 6 Always compare schemes at equal E_b/N_0 , never at equal E_s/N_0 .

References and further reading

- ① S. Haykin, *Communication Systems*, 4th/5th ed., Wiley — chapters on passband data transmission and signal-space analysis.
- ② S. Haykin and M. Moher, *Introduction to Analog and Digital Communications*, Wiley.
- ③ J. G. Proakis and M. Salehi, *Digital Communications*, McGraw-Hill — M-ary signalling and error performance.
- ④ B. P. Lathi and Z. Ding, *Modern Digital and Analog Communication Systems*, Oxford University Press.
- ⑤ T. Pratt, C. Bostian and J. Allnutt, *Satellite Communications*, Wiley — the QPSK link-budget example.

Questions?